Ligjerata 9e

Ekuacioni i gjendjes së gazit ideal

$$p = \frac{1}{3} n_0 m v_m^2 \quad / \cdot \frac{2}{2}$$

$$p = \frac{2}{3} n_0 \frac{m v_m^2}{2}$$

$$p = \frac{2}{3} n_0 E_k$$

$$E_{tr} = \frac{3}{2} \frac{R}{N_A} \cdot T$$

$$p = \frac{\cancel{2}}{\cancel{3}} n_0 \frac{\cancel{3}}{\cancel{2}} \frac{R}{N_A} \cdot T$$

$$n_0 = \frac{N}{V}$$

$$p = \frac{N}{V} \cdot \frac{R}{N_A} \cdot T$$

$$\frac{N}{N_A} = n$$

$$p = \frac{nRT}{V} \quad / \cdot V \quad \Rightarrow \quad \boxed{p \cdot V = nRT}$$ Ek i Klapeyronit

$$\frac{pV}{T} = nR = konst$$

$$\frac{Pa \cdot m^3}{mol K} = \frac{\frac{N}{\cancel{m^2}} \cdot m^{\cancel{3}}}{mol \cdot K} = \frac{J}{mol K}$$

$n = 1\,mol$

$$R = \frac{101325\,Pa \cdot 22{,}41 \cdot 10^{-3} \frac{m^3}{mol}}{273{,}16\,K} = 8{,}314 \frac{J}{mol K}$$